Dynamics of Masked Image Modeling in Hyperspectral Image Classification

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***Abstract*— Masked image modeling (MIM), a common self-supervised learning (SSL) technique, has been extensively studied for remote sensing (RS) image processing. Nevertheless, its effectiveness for hyperspectral imagery (HSI) remains underexplored due to the distinct data structures and high dimensionality. This article aims to provide a detailed understanding of MIM from different perspectives of representation learning and statistical analysis for HSI classification tasks. Our study reveals that the MIM paradigm injects inductive bias in the attention mechanism of the transformer model, which is advantageous for capturing the local discrepancies between the spectra. We also show that MIM can increase the diversity of the attention heads in every layer, which is beneficial for the model in extracting more discriminative features from different spectral bands. The similarity of representations from various layers further proves this. Furthermore, our investigation highlights how MIM introduces a dynamic perspective to spectral representations, enabling the model to learn more robust and discriminative features. The final numerical experiments indicate that a moderate mask ratio can enhance the performance of downstream tasks. This suggests that designing a more targeted masking strategy might be necessary to achieve higher and more stable gains in downstream task performance. Without bells and whistles, the vanilla MIM improves the overall classification accuracy by an average of 2.69% over its supervised learning (SL) counterpart. We hope that our findings can advance the understanding of MIM in HSI and inspire the design of a more stable SSL paradigm for HSI processing.**

***Index Terms*— Hyperspectral imagery (HSI), masked image modeling (MIM), representation learning, self-supervised learning (SSL).**

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I. Introduction

SELF-SUPERVISED learning (SSL) and pretraining techniques are increasingly thinning the attention of the deep learning (DL) community, especially in fine-tuning tasks combined with supervised counterparts. Early DL has been dominated by supervised learning (SL), which highly relies on labeled data. Meanwhile, the success of the SL paradigm has led to significant advances in remote sensing (RS) data interpretation [1]. These advancements have profoundly influenced a wide range of downstream tasks in RS, including but not limited to land cover classification [2], [3], object detection [4], [5], environmental mapping [6], [7], climate analysis [8], etc. However, collecting and annotating labeled data are expensive and time-consuming, especially in the RS community. Recently, access to RS data has been significantly increased, mainly due to the development of satellite and airborne sensors. On the contrary, the number of labeled benchmark datasets accompanying the RS data volume surge trend did not show the same trend [9], [10]. This phenomenon is more severe in the hyperspectral community, where expensive acquisition costs and low spatial resolution make assigning precise categories for every pixel of hyperspectral imagery (HSI) more difficult.

In contrast, SSL methods significantly reduce the dependence on labels, allowing for generic representations from unlabeled data that brings benefits to RS downstream tasks [11], [12]. For instance, Hong et al. [13] proposed an auto-encoder-based EGU-Net for HSI unmixing. Zhu et al. [14] utilized the self-supervised unrolling network to learn a robust mapping from spaceborne imagery to achieve HSI fusion. Chen et al. [15] learned universal representations for HSI target detection in an SSL-based spatial–spectral masking manner. Wang et al. [16] generated the noisy-clean training pairs by using SSL scheme for HSI denoising. Jian et al. [17] first applied the graph SSL contrastive network to maximize the mutual information between feature representations to achieve HSI change detection. The aforementioned evidences show that SSL can learn task-agnostic representations from unlabeled data, which is more cost-effective and scalable. The high quality of the learned representations from SSL methods depends on the appropriate *pretext task* design.[1]

[1]Pretext tasks are designed to teach the model to understand the data structure by predicting certain aspects of it, for example, completing a sentence in textual data or filling in missing regions in the image.

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Based on the pretext task, the category of SSL can be divided into two learning paradigms: contrastive learning (CL) and reconstruction-based learning (RBL) [18].

For the CL part, the core idea is to maximize the similarity between positive pairs and minimize the similarity between negative pairs [19]. The CL paradigm has been widely used in vision representation learning, and it has been applied in various RS vision downstream tasks, such as image classification [20], [21] object detection [22], and semantic segmentation [23]. To adapt CL to the HSI processing, spectral–spatial CL has been proposed [24]. Hou et al. [25] used data augmentation to construct positive and noised negative pairs to improve the performance of HSI classification. Liu et al. [26] used two classes of a few labeled samples to generate contrast group pairs and then used CL to learn the representation of HSI. Similarly, Ou et al. [27] also used Gaussian noised pairs to reduce the influence of the noise in the feature extraction and finally adapted to change detection tasks. Li et al. [28] proposed a guided group CL architecture to enhance the correlation between labeled data and unsupervised counterparts.

Among the various CL-based SSL paradigms, a generative-based method named masked image modeling (MIM) [29] has been widely used in vision representation learning. Different from CL, MIM is an RBL paradigm inspired by masked language modeling [30] in processing text data. The core idea of MIM is to reconstruct the original image from the masked image, and the objective is to minimize the difference between the original image and the reconstructed result. Recently, numerous studies have explored the efficacy of MIM in processing HSI. For example, Ibañez et al. [31] proposed a masked auto-encoder (MAE) for HSI representation learning, and they found that the MAE could alleviate the intrinsic noise of HSI. Scheibenreif et al. [32] proposed a block-wise patch embedding method for mask pretraining, and a spatial–spectral transformer was used to model the factorized visible patches of HSI. Liu et al. [33] proposed spectral masking to keep spatial information and fine-tune the entire network on the classification task. SS-MAE [34] adopted MAE for multisource RS image classification, and they independently masked spatial and spectral domains of HSI. Other works such as [35] and [36] integrated both the idea of MAE with CL to improve the representation ability of features.

The aforementioned reconstruction-based MIM pretrain models have shown their potential in modeling the representation of unlabeled HSI data and can significantly improve the fine-tuning classification performance. However, the intricacies of the MIM representation learning process in HSI remain underexplored. Even though the statistical properties of photographic images and HSI are not significantly different [37], we elucidate the discontinuity inherent in directly transferring takeaways from optical image analysis to the community of HSI representation as follows.

1) *High Dimensionality*: HSI exhibits an increased complexity compared to color photographic images, primarily due to the high dimensionality of the pixel data. The spectral channels typically cover narrow bands of the electromagnetic spectrum. In conjunction with the few labels, the high dimension of the spectra shows substantial redundancy for classification tasks [38].
2) *High Heterogeneity:* The sensitivity to imaging conditions causes distinctive sources of noise and uncertainty in HSIs, ultimately manifesting as the spatial variability of the spectral signature [39]. This high heterogeneity in HSIs is characterized by varying spectral properties across different materials and conditions.
3) *Sequential Spectral Adaptability:* Typically, the vision transformer (ViT) model employs a distinct approach to handling patchy tokens. For optical data, patchification is usually performed along the spatial dimension. In contrast, HSI data is often patchified across the spectral (or channel) dimension to maintain the inter-dependencies of spectra [40] (except for using dimension reduction techniques such as PCA [41] to align the number of channels with optical images).

To investigate the difference between understanding the model dynamics of the MIM paradigm in HSIs and visual data, we aim to answer the following main questions.

1) Q_1: *What do MIM-based SSL models perceive that their SL counterparts do not?* This question explores the unique insights that MIM-based SSL models can detect, which might be overlooked by models trained via traditional SL methods. Specifically, SSL has achieved significant gains when two paradigms employ the same data pattern and model (ViT model with global receptive field). We assume that the adaptive and dynamic adjustment of parameters in SSL models allows them to focus more acutely on informative features.
2) Q_2: *How do the results derived from MIM-based SSL paradigms influence the performance of downstream applications?* (take the HSI classification task as an example). Typically, the MIM paradigm is divided into reconstruction and fine-tuning stages, each utilizing distinct loss functions. This approach may lead the model to learn continuous invariant features across both phases. Consequently, understanding how this affects performance will enhance our insight into the impact of pretraining quality on the learning outcomes of subsequent tasks and overall model performance.

To answer these questions and explicitly delineate the distinctions and inherent correlations between the two paradigms, we compared the MIM model and its supervised counterpart from the perspectives of learnable model parameters and learned features. Additionally, we treat the reconstruction paradigm within MIM as a task with a reference image and select a series of image quality indicators to delineate their relationship with fine-tuned downstream tasks.

The rest of the article is organized as follows. Section II explains the basic principle of MIM and reviews related works on understanding the MIM paradigm. Section II later motivates Section III to design an explanation protocol. In addition, this section introduces the baseline frameworks of comparative approaches. Section IV describes the datasets and basic setting of experiments and provides a detailed analysis of the results. Finally, Section V concludes by discussing the comparative

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results, pointing out some limitations of this study, and suggesting directions for future work.

II. Preliminary and Related Works

A. Background of MIM

Inspired by masked language modeling [30] in processing text data, the reconstruction-based MIM paradigm demonstrates the potential of SSL in vision representation learning. The components of a typical MIM include a mask sample, an encoder, and a decoder. More specifically, we formulate the training procedure for MIM as follows.

Given an input image $X \in \mathbb{R}^{H\times W\times C}$, where H, W, and C are the image's height, width, and channels, respectively. The image is partitioned into N image patches $x \in \mathbb{R}^{N\times \mathrm{PC}}$ determined by the patch size P. After that, the patch mask $M \in \{0, 1\}$ is randomly generated by the predefined mask ratio r. The masked image $x_{\mathrm{vis}} = x \odot M$ is obtained by applying element-wise multiplication between the patch mask and input patches. Finally, the encoder E_θ is used to extract the representation of the masked image, and the decoder D_ϕ is used to reconstruct the original image from the masked representation, where θ and ϕ are the learnable parameters of the encoder and decoder, respectively. The objective of MIM is to minimize the difference between the original image and the reconstructed image.

$$\min_{\theta,\phi} \mathbb{E}_{x\sim X}\mathcal{L}(x \odot (1 - M), D_\phi(E_\theta(x_{\mathrm{vis}}))) \tag{1}$$

where $\mathcal{L}$ is the loss function.

B. MIM for RS Representation Learning

MIM, a widely used paradigm in visual representation learning, has been applied to various vision downstream tasks. Recently, numerous studies have explored the potential of MIM and its specific implementation, the MAE [42] in learning invariant features of different modalities of RS data [1]. For instance, Cong et al. [43] proposed an MAE-based learning scheme with temporal encoding for satellite image representation. Reed et al. [44] and Noman et al. [45] both proposed the scale-aware MAE to learn scale-invariant features from RS images. Nedungadi et al. [46] extended MAE to multiple pretext tasks for multimodal RS representation learning. Hong et al. [47] proposed a 3-D masking strategy to learn the spatial–spectral representation from large unlabeled multispectral images. A similar 3-D MIM scheme could also be found in HSI representation learning [48], [49].

While MIM has achieved remarkable success in RS representation learning, there is still a lack of analysis on why this paradigm is effective for RS data, specifically for HSIs. Therefore, our research aims to bridge this gap by investigating the underlying mechanisms contributing to MIM's effectiveness in HSI interpretation.

C. Analysis and Understanding of MIM

Recently, many studies have revealed the secrets of MIM by analyzing the learned representations through theoretical deduction or empirical investigations. For example, Cao et al. [50] first investigated mathematical understanding of MAE from the operator theoretic and integral equation perspectives. Zhang et al. [51] built a connection between MIM and CL, indicating that MIM implicitly aligned the masked region with the unmasked one. Kong et al. [52] modeled the data generation process of MAE as a hierarchical latent variable architecture. Based on this assumption, they showed how the mask ratio and patch size affect the representation of information. Pan et al. [53] analyzed the mask-reconstruction pretraining (MRP) based on multiview data samples, and they found that MRP can capture the discriminative features in the pretraining stage. Kong and Zhang [54] simplified the MIM formulation into a siamese form, showing that MIM implicitly learns the occlusion-invariant representations. In addition to the above solid theoretical guarantees, some studies highlight superiority of the MIM by comparing or summarizing experimental phenomena. For instance, Xie et al. [55], [56] uncovered the representation capability of MIM from data scale and feature level via extensive controlled experiments. Li et al. [57] highlighted the importance of middle-order interactions in architecture-agnostic MIM. Zhang and Shen [58] empirically studied the latent representations of MIM by disentangling mixed input. Their i-MAE model proved the linear separability and degree of semantics.

Unlike the studies above, which primarily focus on optical data, the representation learning process of MIM in HSI remains ambiguous despite several studies having already applied MIM to HSI processing.

III. Methodology and Protocol

This section comprehensively describes the frameworks of MIM-based SSL and SL baseline methods, ensuring that the comparative approaches maintain consistent parameters and learning strategies in downstream tasks. Section III-B then presents a detailed discussion of the evaluation protocols and the selection of quality metrics, specifically designed to address two principal questions: What do MIM-based SSL models perceive that their SL counterparts do not? And how the results derived from MIM-based SSL paradigms influence the performance of downstream applications?

A. Architecture Specifications

To model the spectral representations without sacrificing the long-range inter-spectral dependencies of HSIs, the ViT model is often a superior choice due to its global receptive field at every layer. Meanwhile, ViT is also a commonly used model in the MIM paradigm. To mitigate the impact of specialized network designs and optimization algorithms on the feature representation, we employ the vanilla ViT model from SpectralFormer (SF) [40] as our baseline supervised method, which is shown in Fig. 1(b). The SF is a pure transformer-based model designed explicitly for HSI processing and has shown its potential in modeling the subtle dependencies of different reflectances. Notably, we do not make any other specific modifications to SF, but we briefly describe its classification process for completeness.

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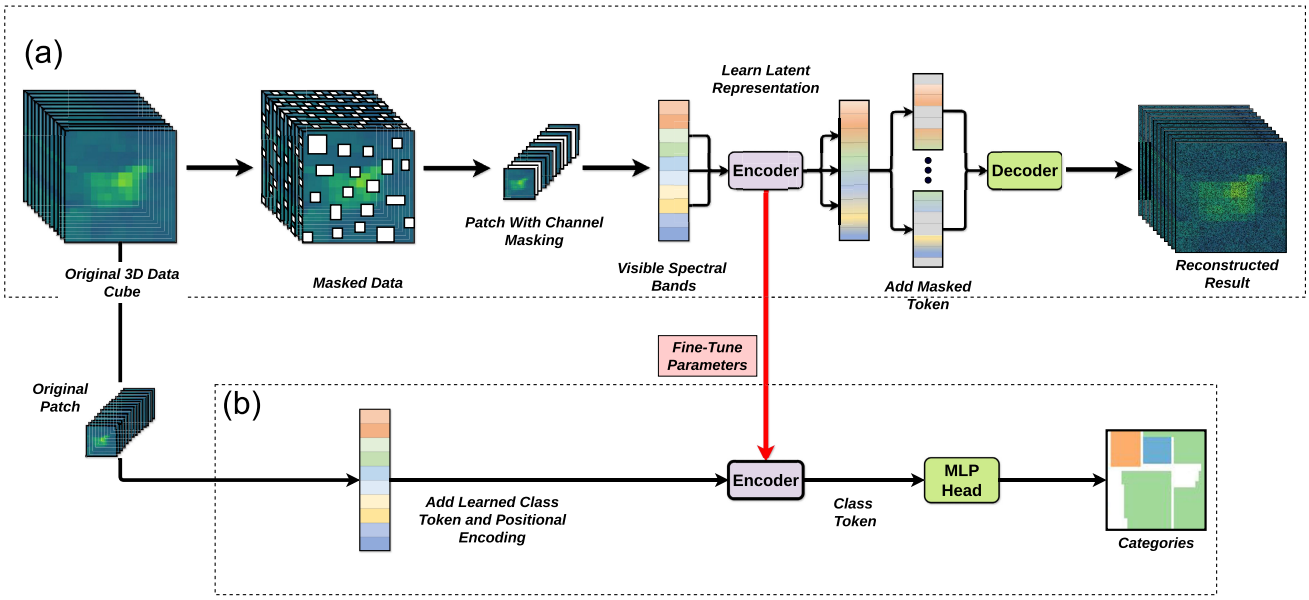


Fig. 1. Overall architecture of the experienced models. (a) Pretext task of MIM-based SSL paradigm and (b) SL paradigm. The connection between these two paradigms is the fine-tuning process after reconstruction pretraining shown in the red arrow.

The overall framework of the SF model can be divided into three parts: data preprocessing, feature extraction, and classification. For the data preprocessing part, the whole HSI cube $X \in \mathbb{R}^{W\times H\times c}$ is spatially disjointed into a series of 3-D patches $x \in \mathbb{R}^{w\times h\times c}$, and each patch is flattened in the spatial dimension to form a 2-D map $\overline{x} \in \mathbb{R}^{wh\times c}$. Then $\overline{x}$ is pretokenized into discrete tokens by projecting it linearly into a latent space $\overline{x} \in \mathbb{R}^{d\times c}$, where d represents the dimension of each token and c represents the number of tokens which is equal to the spectral dimension. After that, the latent tokens are fed into the transformer encoder with learnable positional encoding and class token, which consists of multiple layers of SA and feed-forward networks (FFNs). The output of the transformer encoder is a sequence of spectral tokens, and the included class token is then fed into a linear classifier to predict the class label determined by the center pixel of the original patches. This architecture preserves the integrity of the spectral sequence during feature extraction, enabling the SF model to capture subtle discrepancies more effectively than conventional neural networks.

For the SSL counterpart, we chose MAEST [31], a brilliant successor of SF. As shown in the whole architecture of Fig. 1, the MAEST model comprises two parts. (a) Is a reconstruction-based masked auto-encoder (MAE), which consists of an encoder and a decoder. The encoder is used to extract the representation of the masked image, and the decoder is used to reconstruct the original image from the masked representation. During this process, the MAE model is trained in a self-supervised manner, and the learned parameters of the encoder are then used to fine-tune the downstream classification task. (b) Shares the same tokenization and network architecture as SF, however, under the MIM scheme, apart from the addition of a class token, positional encoding, and a classification head, the remaining learnable parameters are initialized and fine-tuned from the encoder described in the red arrow of Fig. 1.

To preserve ViT and MIM's original representation capabilities, we avoid incorporating supplementary components or employing additional training tricks, such as group-wise spectral embedding methods or cross-layer adaptive fusion strategies mentioned in the literature. Likewise, we followed the standard fine-tuning protocol for the encoder, adjusting the final classification layer and resizing the positional embeddings using bicubic interpolation. The remaining parameters were directly inherited as initialization for the SL paradigm.

B. Evaluation Protocols and Metrics

1) Analysis of $\mathbf{Q_1}$*:* The SF and the MAEST both inherit the global receptive field of the ViT model, which is beneficial for capturing the long-range dependencies of the spectra. Notably, the global perspective is derived from the self-attention mechanism, as described in 2. This mechanism allows each input spectral band to access all other bands within the spectral sequence, while using learnable weights to assess the interrelations between these bands.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V = A \cdot V \tag{2}$$

where Q, K, and V are the query, key, and value matrices, respectively, and d_k is the dimension of the key. The attention matrix A is the dot product between the query and key matrices, followed by a softmax operation. The final output is obtained by multiplying the attention matrix with the value matrix. To address Q_1, we utilize the visualization techniques from [56] to reveal the characteristics of the attention layers. This involves comparing the learnable attention parameters and feature maps generated by the two methods.

For the learnable parameters, we first analyze the attention distance of the SA layer in the ViT model. The reason is that optical images exhibit strong spatial locality between the center pixel and their nearby pixels, which is beneficial for distinguishing the foreground and background. Similarly,

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the continuous form of spectra of HSIs also exhibits strong locality, which contains the subtle discrepancies of different materials that are beneficial for fine-grained classification. However, the vanilla transformer naturally lacks locality inductive bias compared to its CNN counterparts. Therefore, we investigate the locality inductive bias by analyzing the attention distance of the model.

Specifically, we calculate the average distance of the attention weights in each head of the SA layer to examine whether the MIM paradigm can inject the locality inductive bias. The attention distance is calculated as follows:

$$\text{Attention Distance}(x_i, x_j) = A_{i,j} \times D_{i,j} \tag{3}$$

$$D = \begin{pmatrix} 0 & 1 & 2 & \cdots & n \\ 1 & 0 & 1 & \cdots & n-1 \\ 2 & 1 & 0 & \cdots & n-2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n & n-1 & n-2 & \cdots & 0 \end{pmatrix}_{i,j} \tag{4}$$

where $D_{i,j}$ is the distance matrix between the ith and jth spectral band. The distance between each token typically determines the distance matrix D. For optical images, after tokenization in the spatial dimension, the patch size determines the distance between tokens (e.g., in ViT-16 or ViT-32). Since we tokenize in the spectral dimension, with each spectral band serving as a token, the distance between each token is consistently 1. If multihead attention is used, the attention distance is calculated for each head of the SA layer, and the average distance is calculated across all heads.

Second, we analyze the attention distribution of each head by calculating the head entropy. In information theory, entropy is termed as a measurement of the uncertainty of the system and indicates the level of disorder [59]. In our case, a higher entropy value signifies more heterogeneous parameters, suggesting that this head focuses on a broader range of tokens. On the contrary, a head exhibiting lower entropy denotes increased homogeneity of parameters, implying focused attention on a limited number of tokens. The attention entropy is calculated as follows:

$$H = -\sum_{i=1}^{N}\sum_{j=1}^{M} A_{i,j} \log(A_{i,j} + \epsilon) \tag{5}$$

where $A_{i,j}$ is the attention weight of the ith token in the jth head, N is the number of tokens, M is the number of heads, and ϵ is a small constant to prevent the logarithm from approaching infinity.

Finally, further investigation of the degree of diversity among different attention heads regarding the similarity of attention across all tokens was conducted. We measured the Kullback–Leibler (K–L) divergence between the attention maps of other heads. Typically, a lower K–L divergence value indicates that the other heads overly focus on similar features, which may lead to the parameter redundancy of the model. Conversely, diversified attention maps provide different perspectives for modeling the data. The K–L divergence is calculated as follows:

$$D_{\text{KL}}(P \parallel Q) = \sum_{i} P(i) \log \frac{P(i)}{Q(i)} \tag{6}$$

where P and Q are the attention maps of two different heads in the same layer. i represents the index of the element in the attention map.

The analysis of learnable parameters implicitly reflects the network's capacity to model the spectral information. Based on this, we further explicitly analyze the feature maps generated by the attention layers. Specifically, we first calculate the centered kernel alignment (CKA) similarity [60] between the feature representation of each layer. The CKA is calculated as follows:

$$\text{CKA}(K, L) = \frac{\text{tr(KHLH)}}{\sqrt{\text{tr(KHK)} \cdot \text{tr(LHL)}}} \tag{7}$$

where K and L represent the kernel matrices of the feature outputs from two layers. These kernel matrices are computed using a predefined kernel function. The matrix H is the centering matrix, defined as $H = I - (1/n)\mathbf{1}\mathbf{1}^T$ where I is the identity matrix, $\mathbf{1}$ is a vector of ones, and n is the number of data points (feature vectors). This centering matrix ensures that the mean influence of the features is removed, making the similarity measure invariant to shifts in the data mean. The trace operator $\text{tr}(\cdot)$ calculates the sum of the diagonal elements of a matrix. The CKA value ranges from 0 to 1, where 0 indicates no similarity between the feature representations.

2) Analysis of $\mathbf{Q_2}$*:* To better understand the impact of the MIM paradigm on HSI classification, we built an intermediary to explore how masking influences learned representations and, ultimately, downstream task accuracy. To achieve this, we systematically investigate the effect of the masking ratio on reconstruction quality. Since reconstruction quality reflects the strength of learned representations, analyzing this relationship provides valuable insights into the model's performance. The masking ratio is the most critical data-agnostic parameter in MIM, as it directly controls how much information is visible to the network during reconstruction. Therefore, understanding its impact is essential for optimizing downstream task accuracy during fine-tuning. Another key factor influencing reconstruction quality is the scale of the pretraining dataset, which directly impacts the learned representations. In this study, we explore the effect of the data scale on reconstruction quality. However, as this factor is not intrinsic to MIM, the experiments and analysis are provided in Appendix A.

Our study builds on previous findings that consistently show high-quality reconstructions do not always lead to improved performance. They can sometimes degrade downstream task accuracy, as several studies have shown [61], [62]. To further explore these dynamics, we evaluate high-level and low-level representations across spatial and spectral dimensions by comparing the original inputs with their reconstructed counterparts. In the spatial domain, we use the structural similarity index measure (SSIM) [63] and the feature similarity index measure (FSIM) [64] for high-level evaluation, whereas peak signal-to-noise ratio (PSNR) and root mean square error (RMSE) are applied for low-level analysis. Similarly, in the spectral

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TABLE I
LAND-COVER CLASSES OF THE TOULOUSE DATASET

Impermeable						Permeable					
Class No.	**Class Name**	**Unlabeled**	**Train**	**Val**	**Test**	**Class No.**	**Class Name**	**Unlabeled**	**Train**	**Val**	**Test**
1	Orange tile	59140	12215	10085	41888	17	Synthetic grass	24004	7940	7364	29484
2	Dark tile	996	730	820	1742	18	Synthetic track	36640	40097	36805	64026
3	Slate	-	2653	3019	13584	19	Healthy grass	38971	11670	10156	37339
4	Clear fiber cement	4163	9814	14584	18743	20	Stressed grass	41458	9732	9671	32635
5	Dark fiber cement	31623	7247	8388	25006	21	Tree	59746	9663	12290	54549
6	Dark sheet metal	30920	8867	7419	28578	22	Bare soil	91127	27694	25357	80606
7	Clear painted sheet metal	39358	7577	8426	30463	23	Gravels	27241	6199	4892	19924
8	Dark asphalt	40396	7685	7505	34358	24	Rocks	19073	8912	8599	21616
9	Red asphalt	10789	2108	2082	7677	25	Green porous concrete	2581	4039	4330	4042
10	Beige asphalt	119	462	704	1899	26	Porous concrete	3365	2862	3602	4019
11	Green-painted asphalt	4198	854	1047	3989	27	Seaweed	6241	1834	2026	5627
12	White road marking	4341	1562	1301	4180	28	Water - swimming pool	1327	2102	2734	29437
13	Cement	21041	4722	7046	22431	29	Water	31644	12308	19439	37217
14	Brown paving stone	11074	6256	7959	17519	30	Sorgho	-	35755	33245	73600
15	Clear paving stone	13349	3954	4582	15635	31	Wheat	338666	54522	54836	252525
16	Clear plastic cover	3815	4434	4851	8276	32	Field bean	45046	60812	73207	201138
	Total	275322	81140	89818	275968		**Total**	767130	296141	308553	947784
	Combined Total: Pre-Train: 1042452, Fine-Tune: 377281, Val: 398371, Test: 1223752										

domain, the spectral angle mapper (SAM) and spectral information divergence (SID) are used to assess the respective representations [65].

IV. EXPERIMENTAL ANALYSIS ON DOWNSTREAM TASKS

In this section, we conduct experiments to compare the fine-tuning performance of the MIM-based SSL and SL methods on the classification tasks. We first present a comprehensive description of the dataset used in our experiments, detailing its intrinsic characteristics [such as spectral resolution and ground sampling distance (GSD)] and its artificially determined attributes (including data split distribution and sampling strategies) in part A. Next, we show the experimental details, including the pretraining and fine-tuning. Then, we analyze the experimental results from the perspectives of learnable parameters and learned feature representations. Finally, we conduct a statistical analysis to investigate the influence of the pretraining process on downstream tasks.

A. Data Description

Unlike optical data, which can provide a substantial amount of labeled data, hyperspectral data often suffers from a scarcity of labeled data due to the challenges in manual annotation, which is typically restricted to a single scene. When considering spatial information, special attention must be paid to the issue of data leakage. Given these constraints, we chose four different hyperspectral datasets. The Toulouse dataset, with its ample labeled data, will be primarily showcased through pixel-level sampling experimental results. The other three datasets utilized disjointed sampling and incorporated spatial information to compute metrics considering spatial structural information. The detailed descriptions of the datasets are as follows.

1) Toulouse Dataset: The Toulouse (TL) dataset [66], [67] captured over the city of Toulouse, France, using the AisaFENIX 1K camera, offers a spectral range from 0.4 to 2.5 μm. It provides a spectral resolution of 3.6 nm in the VNIR and 7.8 nm in the SWIR, with each pixel comprising 310 bands. The dataset also includes a swath width of 1024 m and a GSD of 1 m. Specifically, for the classification task, the authors utilized three on-ground ASD spectrometers with the same spectral range as the camera to serve as the in situ reference. This data helped define 32 land cover classes, further divided into 16 impermeable and 16 permeable materials. Following the standard split protocol, eight spatially disjoint areas were divided into a training set, validation set, test set, and unlabeled set, containing 13%, 14%, 46%, and 27% of the labeled samples, respectively. The authors further categorized the samples into labeled and unlabeled pools within the unlabeled set. The labeled pool was attributed to specific categories, whereas the categorization of the unlabeled pool remained uncertain. Consequently, the labeled pool was utilized as the pretraining set for SSL. Given the aforementioned data split, the TL dataset is a suitable benchmark for evaluating the performance of SSL methods and their SL counterparts. Table I provides a detailed description of the land cover classes in the TL dataset.

2) Indian Pines Dataset: The Indian Pines (IP) dataset was captured by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor, which covers agricultural areas in northwestern Indiana and consists of 145 $\times$ 145 pixels at a GSD of 20 m. After removing 20 noisy and water absorption bands, 200 spectral bands remain in the 400–2500 nm range with 10-m spectral resolution. A total of 16 land cover classes were labeled in this scene, and the training and testing samples are detailed in the first column of Table II.

3) Pavia University Dataset: The Pavia University (PU) dataset was acquired using the ROSIS sensor, which comprises 610 $\times$ 340 pixels with a GSD of 1.3 m. This dataset includes 103 spectral bands after removing the water absorption bands. All labeled pixels are categorized into nine land cover classes, and the training and testing samples are detailed in the second column of Table II.

4) Houston (HU) Dataset: The HU dataset was collected by the ITRES CASI-1500 sensor over the University of Houston campus and its surrounding urban area. It contains 349 $\times$ 1905 pixels with a GSD of 2.5 m. This dataset encompasses 144 spectral bands after excluding the water absorption bands. All labeled pixels are segmented into 15 land cover classes, with the distribution of training and testing samples provided in the third column of Table II.

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TABLE II
LAND-COVER CLASSES OF MULTIPLE DATASETS WITH STANDARD TRAINING AND TESTING SETS

Datasets	Indian Pines (IP)		Pavia University (PU)		Houston2013 (HU)	
Class No.	Class Name	Training / Testing	Class Name	Training / Testing	Class Name	Training / Testing
1	Corn Notill	50 / 1384	Asphalt	548 / 6304	Healthy Grass	198 / 1053
2	Corn Mintill	50 / 784	Meadows	540 / 18146	Stressed Grass	190 / 1064
3	Corn	50 / 184	Gravel	592 / 1815	Synthetic Grass	192 / 505
4	Grass Pasture	50 / 447	Trees	524 / 2912	Tree	188 / 1056
5	Grass Trees	50 / 697	Metal Sheets	265 / 1113	Soil	186 / 1056
6	Hay Windrowed	50 / 439	Bare Soil	532 / 4572	Water	182 / 143
7	Soybean Notill	50 / 918	Bitumen	375 / 981	Residential	196 / 1072
8	Soybean Mintill	50 / 2418	Bricks	514 / 3364	Commercial	191 / 1053
9	Soybean Clean	50 / 564	Shadows	231 / 795	Road	193 / 1059
10	Wheat	50 / 162	-	-	Highway	191 / 1036
11	Woods	50 / 1244	-	-	Railway	181 / 1054
12	Buildings Grass Trees Drives	50 / 330	-	-	Parking Lot1	192 / 1041
13	Stone Steel Towers	50 / 45	-	-	Parking Lot2	184 / 285
14	Alfalfa	15 / 39	-	-	Tennis Court	181 / 247
15	Grass Pasture Mowed	15 / 11	-	-	Running Track	187 / 473
16	Oats	15 / 5	-	-	-	-
	Total	695 / 9671	Total	3921 / 40002	Total	2832 / 12197

B. Experimental Details

Unless otherwise specified, all experiments were conducted on the NVIDIA A100 Multiinstance GPU platform, utilizing Python 3 and PyTorch for implementation. The Adam optimizer was used for optimization, with a step learning rate scheduler updating the learning rate every 50 epochs. During the pretraining phase, MSE loss measured the discrepancy between the original input and the reconstructed output. Cross-entropy loss was applied during fine-tuning and SL phases to compute classification loss. Each experiment was repeated with five random seeds, and all results were averaged and reported on the test set.

1) Pretraining Phase: To ensure the quality of learned representations and their relevance to downstream tasks, the pretraining set was carefully curated to strongly correlate with the training set, minimizing the influence of out-of-distribution data. The pretraining procedure consisted of the following.

1) *TL Dataset Pretraining:* Pixel-level unlabeled data from the TL dataset served as the pretraining set. The model was trained for 20 epochs with a batch size of 128 and an initial learning rate of 5×10^{-5}. We adopted a masking rate of 0.8, as recommended in [66]. Backbone network parameters, including the number of encoder and decoder layers, attention heads, and token dimensions, were aligned with the referenced literature. Instead of reducing spectral bands to a specific dimension via tokenization, we retained all spectral bands, treating each as an individual token to enhance inter-spectral relationship representation.
2) *IP, PU, and HU Datasets Pretraining:* Following [31], we adhered to pretraining protocols for the IP, PU, and HU datasets while removing all additional modules and techniques to showcase the inherent characteristics of the MIM paradigm. The label of the training set was excluded from the pretraining data to maintain protocol consistency. Training was conducted for [500, 500, 260] epochs with a batch size of 64, an initial learning rate of 1×10^{-5}, and a masking rate of 0.75. Backbone parameters mirrored the referenced literature for consistency.

2) Fine-Tuning Phase: The encoder parameters from the pretrained model were utilized as initial weights for fine-tuning. The classification head and tokens were trained from scratch, as outlined in [31]. Fine-tuning was performed as follows.

1) *TL Dataset Fine-Tuning:* The model was fine-tuned for 20 epochs with a batch size of 128.
2) *IP, PU, and HU Datasets Fine-Tuning:* Fine-tuning was conducted for 1000 epochs on the IP dataset, 500 epochs on the PU dataset, and 500 epochs on the HU dataset, all with a batch size of 64.

For all supervised ViT models, identical settings for fine-tuning and model architecture were employed across datasets to ensure experimental consistency.

3) Evaluation Phase:

1) *Pretraining Metrics:* For all datasets, representational-level metrics were calculated exclusively on data used during the pretraining phase. These metrics were computed by comparing the reconstruction results with the original input.
2) *Testing Metrics:* Testing was performed on the fine-tuned model, and all parameter analysis, feature map measurements, and accuracy metrics were calculated on the test sets specified in Tables I and II.

C. From the Perspective of Learnable Parameters

The learnable SA mechanism in the ViT model is key to the success of the feature representation. The SA mechanism captures the long-range dependency of the input data, which is crucial for modeling the inner-spectral dependencies of HSIs. In this part, we compare the learnable attention mechanisms in both paradigms to illustrate the differences in what the SSL perceives compared to the SL.

1) Attention Distance: The average attention distances across various attention heads and layers for four datasets are depicted in Fig. 2. Based on the results, it is evident

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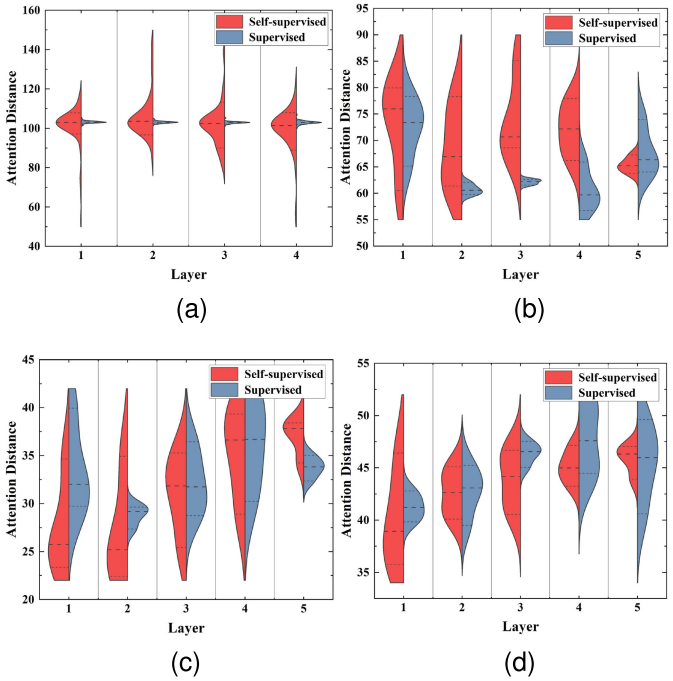


Fig. 2. Average attention distances of different attention heads with different layers on (a) TL, (b) IP, (c) PU, and (d) HU datasets.

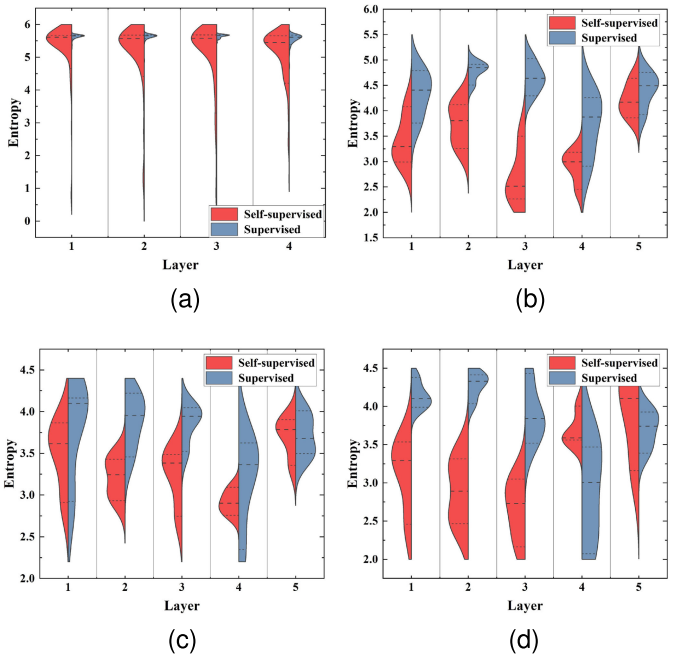


Fig. 3. Entropy of different attention heads with different layers on the (a) TL, (b) IP, (c) PU, and (d) HU datasets.

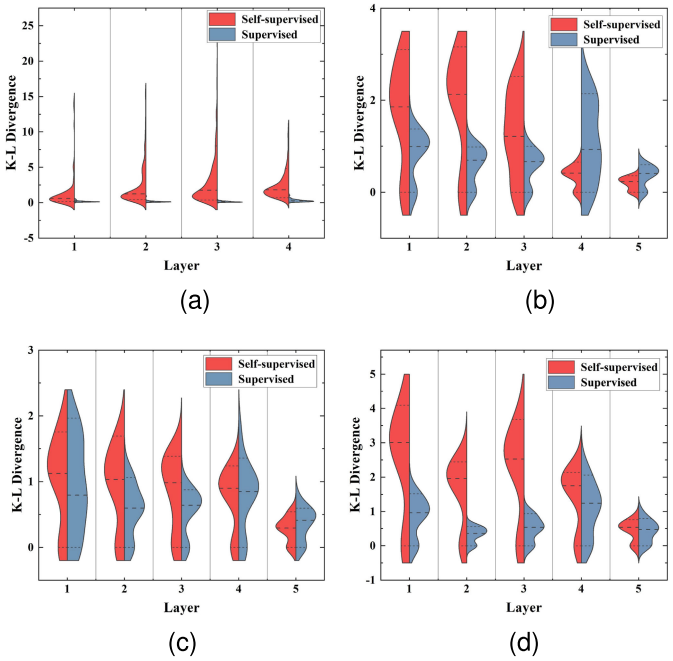


Fig. 4. K–L divergence between every two heads with different layers on the (a) TL, (b) IP, (c) PU, and (d) HU datasets.

that the MIM paradigm introduces greater diversity in the attention heads compared to SL. Particularly in the TL dataset, the MIM paradigm exhibits a significant variety of local and global attention patterns, suggesting an emphasis on learning a mixture of regional and global features across all the stages. In contrast, SL maintains a relatively stable attention space, with minimal differences in attention between the shallow and deep layers of the network (most of the heads focus on approximate attention distance). This is consistent with the inherent characteristics of the ViT model, which ensure that even the shallow layers keep a global receptive field. However, in HSI classification tasks, not all spectral bands are needed. Global spectral information facilitates the modeling of correlations between bands, whereas local spectral information is crucial for identifying subtle discrepancies in spectra. This is particularly useful in mitigating the influence of inter-class similarities and intra-class variability.

These observations show some differences compared to the conclusions derived from optical images. The MIM paradigm reduces the attention distance in the optical image as the network goes deeper. This occurs because the foreground of optical images is often aggregated in localized areas, and the local inductive bias introduced by MIM aggregates the attention on the foreground pixels within the target head. However, for HSI, even after the injection of a local inductive bias by MIM, some heads in the final layer of the network still maintain a relatively wide attention distance. The cause of this phenomenon is the different tokenization methods used. HSIs are tokenized along the spectral dimension, often requiring integrating global and local information during classification. For instance, distinguishing between "green porous concrete" and "red porous concrete" in the TL dataset involves recognizing spectral similarities globally and noting local differences. When differentiating between categories such as "water—swimming pool" and "water," over-focusing on local details may lead to cognitive biases.

Based on the analysis above, the MIM paradigm introduces diverse prediction distances during the fine-tuning stage, enhancing the model's focus on local features without losing its inherent global attention. This further improves the fine-tuning classification performance.

2) Attention Entropy: Entropy is a statistical measure quantifying uncertainty in a probability distribution [68]. As depicted in Section III-B1, entropy in the attention mechanism can be used to measure the concentration of the attention distribution. Specifically, suppose the output weights of an

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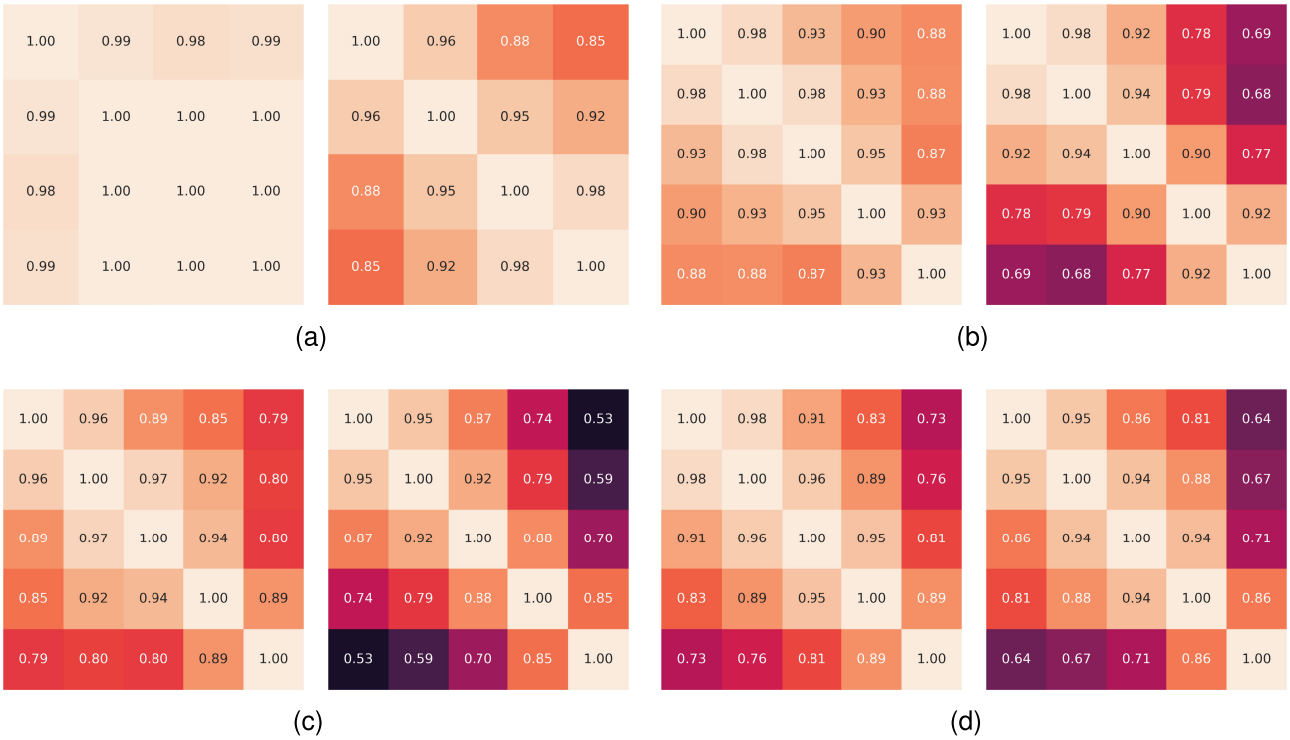


Fig. 5. CKA similarity between the representations generated by each layer of the supervised and self-supervised models is illustrated. Each label corresponds to a different dataset, with (left) SL results and (right) SSL results. (a) TL dataset. (b) IP dataset. (c) PU dataset. (d) HU dataset.

attention head are highly concentrated, with most of the attention focused on a few tokens. In that case, the entropy of this head will be relatively low, indicating a focused and high certainty in its attention allocation. Conversely, suppose the weight distribution is evener, and nearly every token receives similar attention. In that case, the entropy of this head will be higher, indicating a dispersed attention distribution with higher uncertainty.

Based on the experiment results shown in Fig. 3, from a macro perspective, we observed that, compared to SL, except for the last two layers of the HU dataset, the entropy of attention heads in every layer is reduced under the MIM paradigm. The MIM paradigm also demonstrates greater diversity in the distribution of entropy. This suggests that, in each layer, some heads focus on global spectral information, while others concentrate on local spectral details. From a micro perspective, in SL, most heads in the shallow layers exhibit high entropy, showing a broad focus of attention. Gradually, some low-entropy heads emerge in the middle layers to focus on more specific information. However, in MIM, focused attention is apparent from the shallow layers and is further intensified in the middle layers. We then compare Figs. 2(a) and 3(a), which further reveal that, for the middle layers (layers 2 and 3), some heads in MIM paradigm have a higher attention distance than those in SL. However, the entropy for corresponding heads in Fig. 3 shows that the entropy of MIM is almost overall lower than in SL. This suggests that, although the attention covers a wide range of input bands, only a few bands may receive the most attention weights. Similar patterns are also observable when comparing specific layers in the IP and PU datasets (layers 2–4) and the HU dataset (layers 2 and 3).

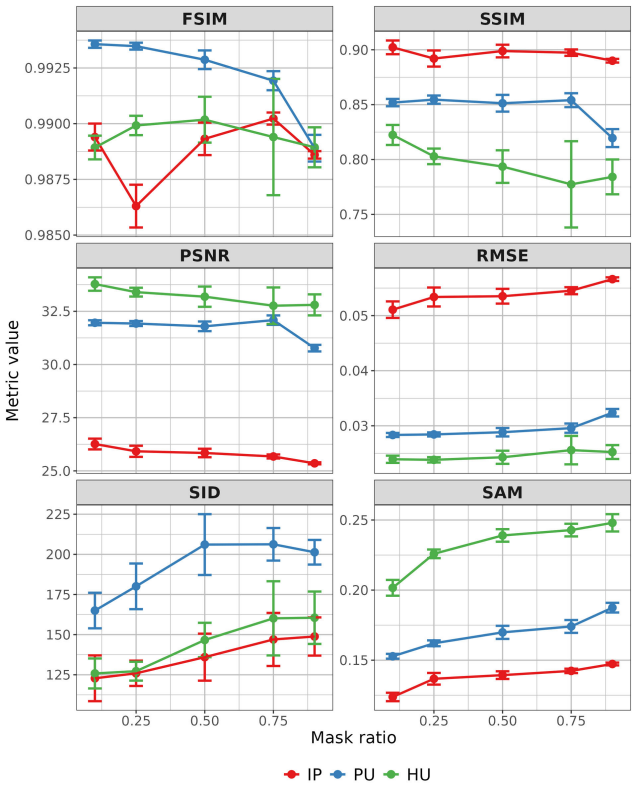


Fig. 6. Assessment of MIM model reconstruction quality using image quality metrics across various mask ratios for IP, PU, and HU datasets: higher scores are better for FSIM, SSIM, and PSNR; and lower scores are better for SID, SAM, and RMSE.

Based on the analysis above, the MIM paradigm introduces certain information about attention heads during the fine-tuning stage, enhancing the model to focus on a few tokens

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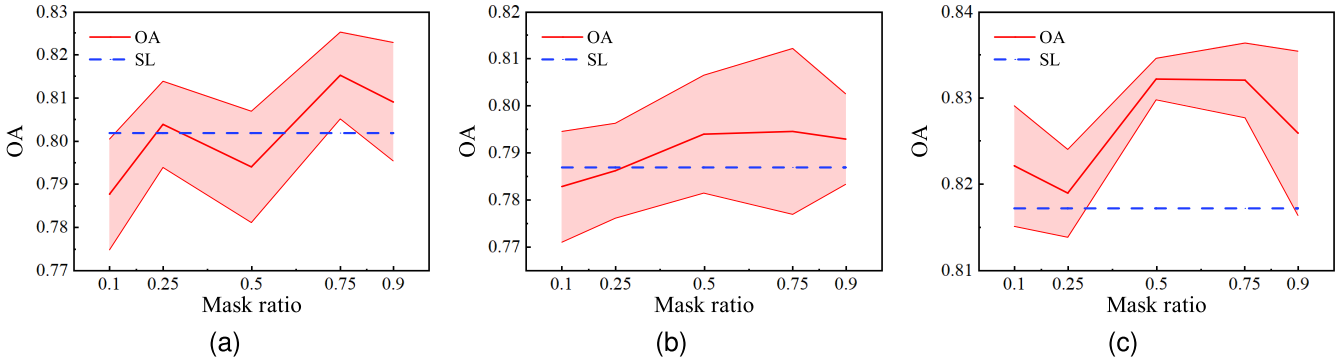


Fig. 7. Overall accuracy of the self-supervised models on (a) IP, (b) PU, and (c) HU datasets for different masking ratios. The blue dashed line is the accuracy of SL.

(spectral bands) to improve classification accuracy. Although middle attention heads can access a broader range of spectral bands, they ultimately focus on only a few specific bands. We can consider that the MIM paradigm is performing a *separate wheat from the chaff* way, with the potential to resist redundancy and noisy spectral bands.

3) Attention Divergence: The previous experimental results demonstrate the MIM paradigm's ability to introduce the model with more diverse attention capabilities. This enables the model to focus not only on global spectral distributions but also on local spectral discrepancies. However, there is no denying that most heads exhibit similar attention distance and entropy behaviors. This could cause the network to excessively focus on similar spectral bands or features when processing spectral information. As a result, there may be redundancy in the parameters. We calculate the K–L divergence between the attention maps of different heads for each layer. This investigation helps determine whether the model focuses on similar or different spectral bands. The results are shown in Fig. 4.

We can observe that the attention divergence under the MIM paradigm is higher than that in SL across nearly all layers, especially in every layer of the TL dataset, where some head's divergence in MIM exceeds by nearly ten times than SL. As expected, the MIM paradigm introduces a more diverse attention distribution across different heads, which benefits the model in focusing on different spectral bands. However, both paradigms exhibit lower divergence in the final layer for the IP, PU, and HU datasets. This may be due to their fewer categories than the TL dataset.

D. From the Perspective of Learned Feature Representation

Analyzing the learnable parameters offers a theoretical understanding of how the MIM paradigm converges and provides insights into how to adapt these parameters to the HS data distribution explicitly. Examining the feature representations offers a more intuitive understanding of how these parameters impact decision-making. To explore this, we compute the CKA similarity between feature representations across different layers for supervised and self-supervised models, as illustrated in Fig. 5.

Our results indicate that, as the network deepens, CKA similarity decreases progressively across all paradigms, suggesting that the backbones learn increasingly distinct features at different layers. Notably, the MIM paradigm exhibits a more rapid decline in similarity, signifying bigger differentiation between learned features across layers (e.g., CKA similarity shifts from the $(0.7, 1]$ range for SL to $(0.5, 1]$ for MIM), reflecting increased feature diversity. This finding further corroborates the conclusions from our earlier analysis of learnable parameters, showing that, by introducing diverse attention, the model captures features related to category labels from multiple perspectives.

The observation also reveals some distinctions compared to previous research on optical images [54], [56]. Notably, it suggests that the MIM paradigm facilitates the learning of occlusion-invariant features and promotes the effective propagation of representations between shallow and deep layers. However, these characteristics may not be directly applicable to HSI classification tasks. The ViT model's unique tokenization process allows it to better capture inter-dependencies between spectral bands. If the model overemphasizes specific spectral ranges or features during material classification, it could become more sensitive to spectral variability. As a result, when MIM introduces a local inductive bias, fine-tuning parameters across different ViT layers begins to exhibit characteristics more akin to CNN architectures [69].

E. From the Perspective of Statistical Metrics

We gradually increased the masking ratio from extremely low value of 0.1 to extremely high of 0.9. This allowed us to observe changes in the quality of reconstructed images. Simultaneously, we fine-tuned corresponding parameters to better understand the relationship between the reconstruction process and the performance of downstream tasks. The change of quality metrics for IP, PU, and HU datasets is shown in Fig. 6, and the corresponding overall accuracy is shown in Fig. 7.

Based on the results, all datasets achieve the highest downstream accuracy gains at moderate masking ratios (e.g., IP and PU achieved optimal results at a masking ratio of 0.75, HU at 0.5, and TL at 0.7 and 0.8, as reported in [66]). An extremely low mask ratio obtains the lowest accuracy (e.g., IP and PU get the worst results at a masking ratio of 0.1, and HU at 0.25), even lower than SL for the IP and PU datasets. However, when we observe the image quality metrics corresponding

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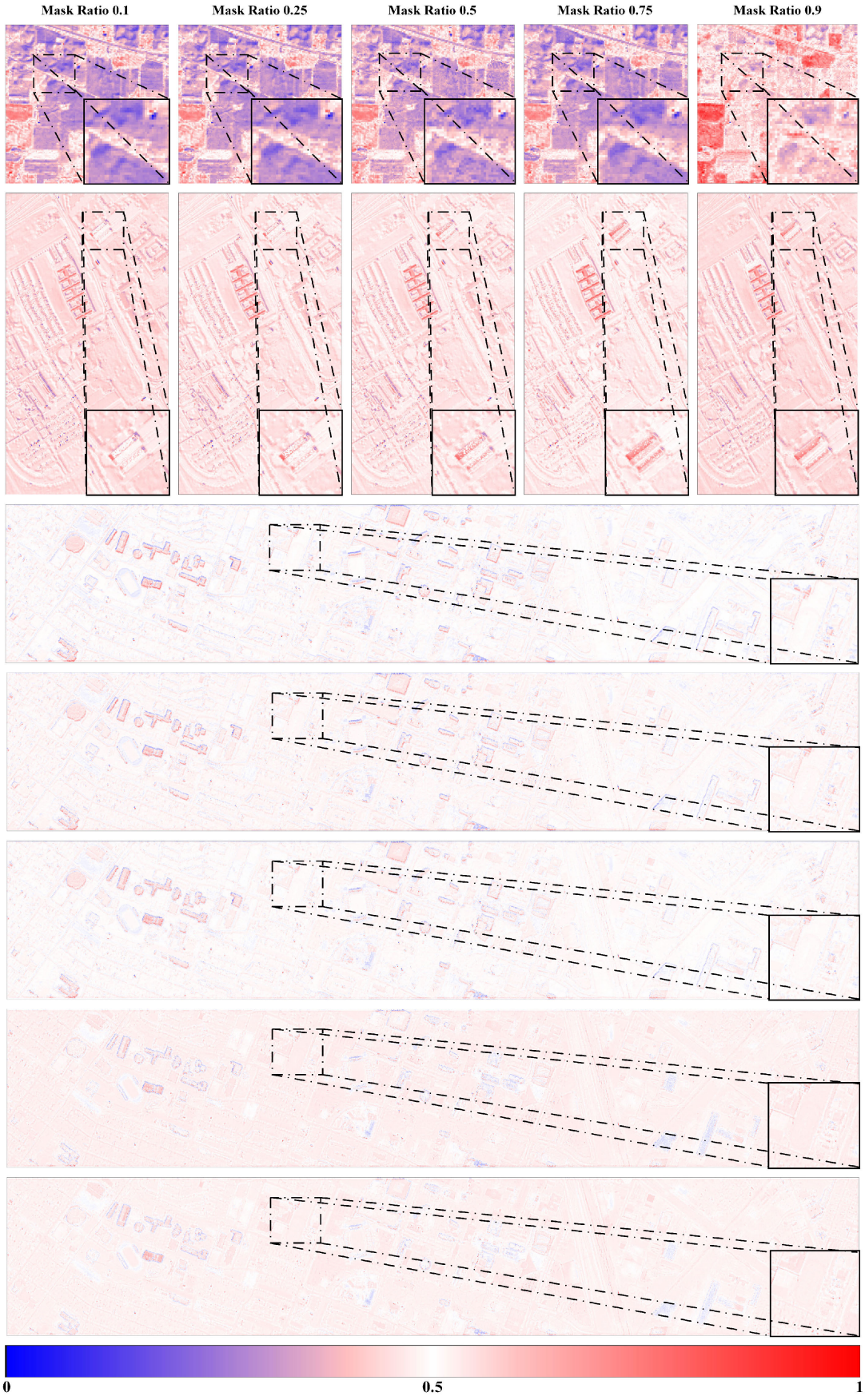


Fig. 8. Reconstruction errors across varying mask ratios. The first row shows the IP dataset, the second shows the PU dataset, and the remaining rows display the HU dataset with increasing mask ratios. The color bar quantifies error values from 0 (blue, minimal error) to 1 (red, maximal error).

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to the mask ratios, we find that nearly all indicators in the spatial and spectral domain decrease as the mask ratio increases (except FSIM of the IP dataset). Specifically, they tend to get the highest reconstruction quality at the lowest mask ratio and the worst at the highest. This phenomenon can be explained because, for the classification task, which is essentially a process of mapping high dimensional data to low dimensional labels, there is a large amount of redundancy in the original HS data for a finite number of categories. When an extremely low mask ratio is taken, the SSL process of the MIM paradigm can reconstruct the masked portion more quickly because it has more input information available to work with. This is easier than reconstructing the input with less information using high mask ratios. However, the original redundant information is retained, and the parameters in the MIM paradigm encoder will inevitably pay too much attention to the redundant information. This can potentially decrease classification accuracy compared to the SL paradigm. This is also why extremely high masking rates do not yield the highest classification accuracy, but they are still much better than extremely low masking ratios.

Another noteworthy phenomenon is that, although higher classification accuracy is achieved at moderate mask ratios, this gain is unstable. Despite using multiple sets of random seeds for training, there is considerable fluctuation in both classification accuracy and reconstruction quality compared to results at extreme mask ratios. This variability may stem from the random masking process used in the MIM paradigm, making it challenging to determine consistently whether the masked information is redundant or essential. The randomness in this process can result in significant variations in the learning process and outcomes, as it depends on which specific data portions are masked during training.

F. Visual Interpretation

To highlight the relationship between reconstruction quality and mask ratio, we visualize the residuals between the original and reconstructed bands using a contrastive blue–white–red colormap. This approach allows us to examine how each dataset responds to varying levels of data masking, focusing on structure preservation and error distribution. For our analysis, we interpret the 41st band of the IP dataset, the first band of the PU dataset, and the 20th band of the HU dataset (it is important to note that the reconstruction process was only trained on the training sets, while the results are inferred from the entire image).

As shown in Fig. 8, the color bar at the bottom of the figure provides a visual scale for interpreting the error values, ranging from 0 (blue representing minimal error) to 1 (red representing maximal error). For the IP dataset, low mask ratios yield predominantly blue regions, indicating minimal reconstruction errors and effective preservation of major structures. The model demonstrates a strong ability to retain the integrity of larger and more distinct features under these conditions. However, increasing mask ratios makes a noticeable shift toward red regions, reflecting significant reconstruction errors. Fine details and smaller structures become less distinguishable, resulting in higher discrepancies, particularly in more complex regions. This effect is even more pronounced in the HU dataset. At higher mask ratios (0.75 and 0.9), the visualizations are dominated by red, indicating that the model struggles to reconstruct areas with detailed and varied terrain, such as complex urban layouts. The errors are particularly severe in regions with intricate structural details, where the model cannot accurately reconstruct the finer components of the image. Across all datasets, more uniform and continuous areas, where pixel values vary less, are better preserved. This suggests that the model can better infer the general characteristics of such regions, even with significant data masking. In contrast, areas with more variability and structural complexity, such as urban or highly textured regions, exhibit larger reconstruction errors as the mask ratio increases.

V. Conclusion

In this work, we investigated how the MIM paradigm performs for HSI classification by comparing it with the supervised paradigm from various perspectives. Then, we conducted a detailed study of the relationship between the reconstruction and fine-tuning stages in the MIM paradigm. From the learnable parameters, we found that the MIM paradigm introduces a more diverse attention distribution across different heads, which introduces the local inductive bias to the model. This improves the backbone in modeling the inter-spectral dependencies of HSIs in diverse receptive fields. After comparing the different characteristics of the attention heads, we further found that the MIM paradigm inherently exhibits properties similar to denoising or de-redundancy. From the learned feature representation, we found that the MIM paradigm introduces a more diverse feature representation across different layers, which further enhances the model's capability to capture features related to the category labels from multiple perspectives. The statistical experiments showed that the MIM paradigm achieves the highest downstream accuracy gains at moderate masking ratios. However, we still need to pay attention to the randomness of the MIM paradigm, which may lead to differences in outcomes.

The limitation of our work is that we only focus on the classification task, and the results may not apply to other tasks such as regression. Due to the lack of sufficient labeled HS data similar to the TL dataset, the training, fine-tuning, and test sets of IP, PU, and HU datasets are from one scene of the image. In our future steps, we will discuss the impact of separately masking the spatial and spectral dimensions to further refine our approach, and then plan to design a more effective masking strategy that leverages MIM's intrinsic band identification properties to achieve more robust classification accuracy. Note that incorporating local inductive biases is necessary when employing the transformer architecture for HSI, and diverse attention mechanisms can significantly enhance classification tasks. Additionally, we will extend our investigations to additional downstream tasks, such as regression and detection tasks. We hope that our findings can deepen our understanding of HSI processing, especially using the MIM paradigm. We also hope our results may inspire more stable and efficient SSL methods for HSI classification tasks.

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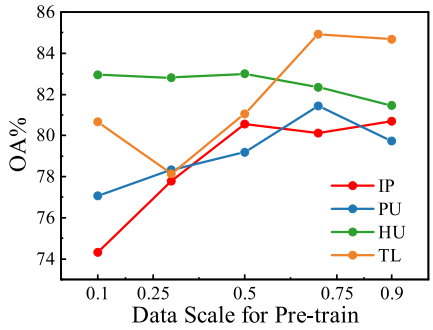


Fig. 9. Impact of varying pretraining data scales on the classification accuracy for four different datasets. Each dataset exhibits distinct trends as the data scale for pretraining increases from 0.1 to 0.9.

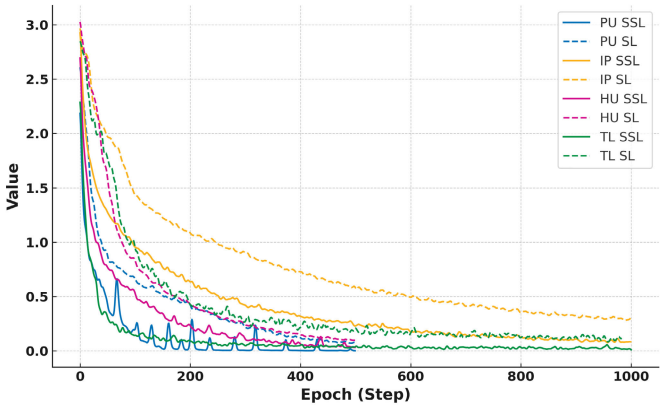


Fig. 10. Convergence rate of SL versus SSL across four datasets (solid line is SSL and dashed line is SL). Since we only trained for two epochs on the TL dataset, we tracked its loss at each step.

APPENDIX A
PRETRAINING DATA SCALE

The amount of data is a more intuitive factor that affects the representation ability of DL models. Fig. 9 illustrates the effect of varying pretraining data scales on classification accuracy (OA%) for four hyperspectral datasets. For the IP dataset (red line), accuracy steadily increases as the data scale grows, peaking at 82% at a scale of 0.75, followed by a slight decline at 0.9, likely due to overfitting. The same trend is observed in the TL dataset (orange line), with accuracy reaching 85% at a scale of 0.75 before decreasing slightly at 0.9. The PU dataset (blue line) also shows an initial sharp increase, with accuracy stabilizing around 80% after a scale of 0.5, indicating diminishing returns from further data increases. The HU dataset (green line) maintains consistently high accuracy across all data scales, showing minimal sensitivity to data size changes.

In summary, while larger pretraining datasets generally improve classification accuracy, the benefits diminish beyond certain thresholds (0.5 or 0.75), with sensitivity varying by dataset. The HU dataset is more robust to data scale changes, while the IP, PU, and TL datasets are more responsive to data scale increases, particularly at lower scales.

APPENDIX B
CONVERGENCE RATE

Fig. 10 compares the convergence behavior of SSL and SL across four datasets. From a convergence perspective, SSL consistently outperforms SL by achieving faster and lower loss values across all datasets. In the PU dataset, SSL rapidly reduces loss within the first 200 epochs, while SL converges more slowly and stabilizes at higher loss levels. A similar trend is observed in the HU and TL datasets, where SSL reaches a lower, smoother loss curve than SL's more gradual convergence. Although SSL converges faster initially in the IP dataset, its loss curve shows more fluctuations, indicating instability. Despite this, SSL achieves a lower loss than SL, though with less stability than in the other datasets.

Overall, SSL demonstrates superior convergence speed and lower final loss than SL across all datasets, particularly in PU, HU, and TL. However, the IP dataset shows that SSL's faster convergence may come at the cost of greater instability, suggesting the need for careful tuning in some cases.

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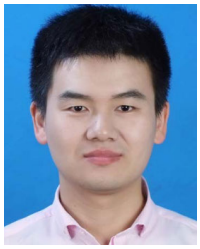
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